Title:-

Prepare aqueous solutions of various solid salts available in the laboratory. Observe what happens when aqueous solution of sodium hydroxide is added to these. Prepare a chart of double displacement reactions based on these observation.

Introduction:-

Chemical reactions are fundamental processes that form the basis of chemistry. Among these, double displacement reactions are particularly significant as they involve the exchange of ions between two reacting compounds, often resulting in the formation of a precipitate, a gas, or a neutral compound such as water.

In this project, we aim to study such reactions by preparing aqueous solutions of various salts and observing their interaction with sodium hydroxide (NaOH). Sodium hydroxide is a strong base that reacts with metal salts to produce metal hydroxides. This experiment provides insights into the solubility of different compounds and the behavior of ions in aqueous solution.

Materials Required:

1. Solid salts:

- Copper sulfate ($CuSO_4$)
- Iron(II) sulfate ($FeSO_4$)
- Zinc sulfate ($ZnSO_4$)
- Lead nitrate ($Pb(NO_3)_2$)
- Calcium chloride ($CaCl_2$)

2. Chemicals and Equipment:

- Sodium hydroxide (NaOH) solution
- Distilled water
- Beakers (5)
- Test tubes and test tube rack
- Glass rod
- White tile (for observing precipitate colors)
- Dropper
- Safety gloves and goggles

Theory:

A double displacement reaction occurs when two ionic compounds in aqueous solutions exchange their positive and negative ions, leading to the formation of new compounds.

The general form of such a reaction is:

$AB + CD \rightarrow AD + CB$

Where:

- AB and CD are reactants (ionic compounds).
- AD and CB are products, one of which may be a precipitate.

The formation of an insoluble compound (precipitate) or a compound with distinct properties (e.g., color) is often the driving force for these reactions. In this experiment, sodium hydroxide reacts with salts to form metal hydroxides, some of which are insoluble and form precipitates.

Procedure:

Preparation of Salt Solutions:

1. Take 5 clean beakers and label them with the names of the salts: $CuSO_4$, $FeSO_4$, $ZnSO_4$, $Pb(NO_3)_2$, and $CaCl_2$.
2. Add 100 mL of distilled water to each beaker.
3. Add a small amount of each salt to the respective beaker and stir until the salt dissolves completely.

Reaction with Sodium Hydroxide:

1. Transfer 10 mL of each salt solution into separate test tubes using a dropper.
2. Add a few drops of sodium hydroxide solution to each test tube, one at a time.
3. Stir gently with a glass rod and observe the changes.
4. Record observations such as color change, formation of precipitates, or any other visible effects.
5. Repeat for all five salt solutions.

Observation table:-

Observations:

Salt Solution	Reaction with NaOH	Observation	Nature of Precipitate
Copper sulfate ($CuSO_4$)	$CuSO_4 + 2NaOH \rightarrow Cu(OH)_2\downarrow + Na_2SO_4$	Blue precipitate forms	Insoluble, blue ($Cu(OH)_2$)
Iron(II) sulfate ($FeSO_4$)	$FeSO_4 + 2NaOH \rightarrow Fe(OH)_2\downarrow + Na_2SO_4$	Dirty green precipitate forms	Insoluble, green ($Fe(OH)_2$)
Zinc sulfate ($ZnSO_4$)	$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2\downarrow + Na_2SO_4$	White precipitate forms	Insoluble, white ($Zn(OH)_2$)

Salt Solution	Reaction with NaOH	Observation	Nature of Precipitate
Lead nitrate ($Pb(NO_3)_2$)	$Pb(NO_3)_2 + 2NaOH \rightarrow Pb(OH)_2\downarrow + 2NaNO_3$	White precipitate forms	Insoluble, white ($Pb(OH)_2$)
Calcium chloride ($CaCl_2$)	$CaCl_2 + 2NaOH \rightarrow$ No visible reaction	No precipitate observed	None

Discussion:

1.Color Change and Precipitate Formation:

The formation of colored or white precipitates indicates a chemical reaction. For instance:

- The blue precipitate in $CuSO_4$ reaction suggests the formation of $Cu(OH)_2$, which is characteristic of copper compounds.
- The dirty green precipitate in $FeSO_4$ reaction confirms the presence of $Fe(OH)_2$.

2. Solubility Trends:

- Calcium hydroxide, though slightly soluble, did not form a noticeable precipitate, indicating its high solubility compared to other metal hydroxides.
- Other metal hydroxides formed insoluble precipitates due to their low solubility product constants.

3. Ionic Exchange:

- The ions of the salt solutions exchange with those of sodium hydroxide, resulting in the formation of insoluble hydroxides.

Advantages:

1. Helps in understanding double displacement reactions clearly.
2. Shows formation of precipitates through visible color changes.
3. Improves practical knowledge of chemical reactions in laboratories.
4. Useful for identifying different metal ions in solutions.
5. Develops observation and experimental skills in students.

Disadvantages:

1. Some chemicals used may be harmful if handled carelessly.
2. Requires proper safety equipment like gloves and goggles.
3. Certain salts and chemicals may be costly or difficult to obtain.
4. Incorrect measurements can affect the observations.
5. Disposal of chemical waste must be done carefully to avoid pollution.

Conclusion:-

The experiment demonstrates the occurrence of double displacement reactions when sodium hydroxide reacts with various salt solutions. These reactions result in the formation of metal hydroxides, some of which are insoluble and form precipitates. The observations validate the principles of ionic exchange and solubility in aqueous solutions.

Precautions:

1. Handle sodium hydroxide carefully as it is highly corrosive.
2. Use safety gloves and goggles during the experiment.
3. Clean all apparatus thoroughly after use.
4. Dispose of chemical waste according to laboratory safety guidelines.

References:

1. NCERT Science Textbook - Chemical Reactions and Equations
2. Laboratory Manual of Chemistry for Secondary Classes
3. www.ncert.nic.in
4. www.britannica.com
5. [www.chemguide.co.uk](http://www.chemguide.co.uk